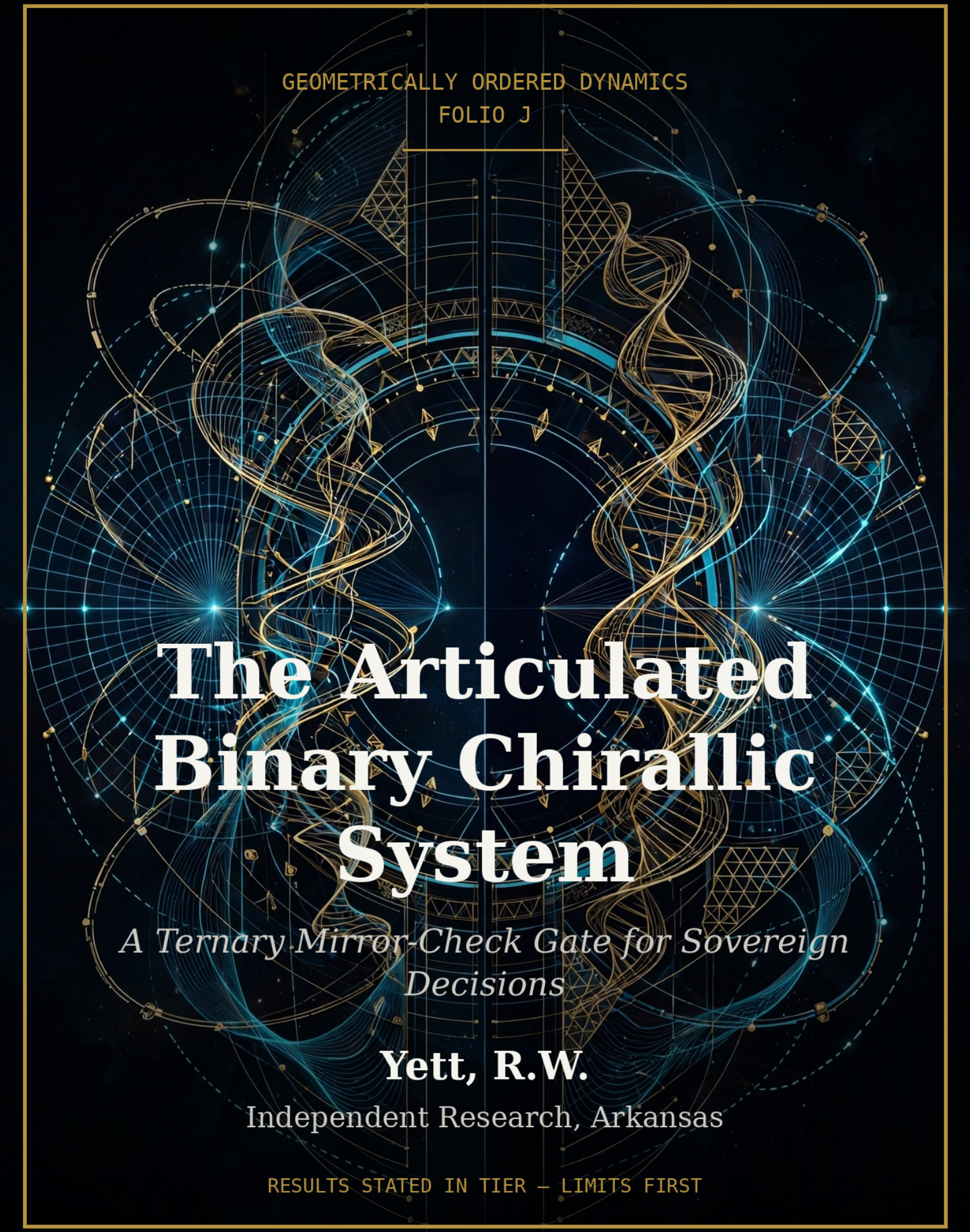
GEOMETRICALLY ORDERED DYNAMICS
FOLIO J



The Articulated Binary Chirallic System

A Ternary Mirror-Check Gate for Sovereign Decisions

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RESULTS STATED IN TIER — LIMITS FIRST

CONVERGENCE -- THE OBSERVER (ARI / sovereign cognition)

This folio converges on **the Observer** read as a compute substrate: the claim that safe autonomous decision-making can be built into the logic gate itself rather than trained into a probability distribution. Its unique angle is the **ternary compute substrate**: decisions as geometric handedness, with execution permitted only when a signal and its chiral inverse are maximally distinguished. Sibling papers approach the same claim from other angles: [TIER: CONJECTURE] the dial folio (IT-COGNITION) reads the Observer as engineered drift control, the network folio (IT-NEURO) reads it as a biological substrate, and the starpolygon folio (IT-OMEGA) reads it as a layered architecture. Limits first: the coherence theorem is specification-level — end-to-end tamper-proofness is conditional on a formalized evaluator, which is the unformalized hard problem; the 49% figure is illustrative, not measured.

SUPERSESSION NOTE

This paper supersedes Zenodo record 20651511 ("The Articulated Binary Chirallic (ABC) System: A Structural Foundation for Sovereign Intelligence," May 2026). Corrections, per audit finding #11: (1) "unconditional. . . tamper-proof" is qualified — the guarantee is conditional on the unformalized eval() (§5); (2) the "49% margin of uncertainty" is clarified as illustrative arithmetic on the arbitrary 0.51 threshold, not a measured quantity (§1); (3) the five truncated bare "Yett (2026)" references are replaced with honest in-corpus citations, the deposited list marked truncated as deposited (§6); (4) the decorative H_{RY}/GOD-holonomy invocations are kept as motivation, not derivation. Orphan check: no Zenodo record supersedes the ABC mechanism, and the Chyren stack's `chyren-ternary` crate implements this gate (starpolygon folio, IT-OMEGA) — this successor is its canonical republication, neither orphaned nor replaced.

1 The problem: probability drift

Standard LLM agents decide by thresholding a softmax distribution: an action executes because a confidence score crossed an arbitrary boundary (e.g. $p > 0.51$). Two structural deficiencies follow. *Threshold arbitrariness:* the boundary is derived from no property of the decision space — a 0.02 shift flips ABORT to EXECUTE with no change in semantic content. *Symmetry blindness:* thresholding has no notion of the decision's logical inverse — nothing verifies that the counter-factual would be rejected with comparable confidence. This asymmetry is the source of hallucinated commitments. **Probability drift** is defined as the failure mode in which a cognitive state S transitions to execution despite lacking a stable chiral inverse.

On the 49% figure. The predecessor's abstract speaks of "a 49% margin of uncertainty in every decision." This figure is *illustrative*, not measured: it is the arithmetic complement $1 - 0.51$ of the arbitrary threshold used as the running example. It is carried here in that explicit capacity — a dramatization of threshold arbitrariness, not an empirical estimate of decision uncertainty.

2 The ternary mapping

Let $X = [-1, 1]$ be the continuous signal space of a cognitive state. The **ternary mapping** $T : X \to \{-1, 0, +1\}$ is

$$T(x) = \begin{cases} +1 & x > \varepsilon \\ -1 & x < -\varepsilon \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

where $\varepsilon > 0$ is a coherence margin defining the dead-band around the symmetry point. The design choice $\varepsilon = 1 - \chi_s \approx 0.046$ anchors the dead-band to the sovereign boundary $\chi_s = 0.9539$: signals scoring above χ_s map to ± 1, signals below are held in the SUSPEND zone. The three values carry the semantics $+1$ (Execute), -1 (Abort), 0 (symmetry collapse / uncertainty).

The **chiral operator** $\chi : \{-1, 0, +1\} \to \{-1, 0, +1\}$ is the sign-inversion map $\chi(s) = -s$, an involution with the single fixed point $s = 0$ — the geometric signature of ambiguity, a signal indistinguishable from its own inverse. **Lex Prima (law of coherence):** a signal s is chirally coherent iff $s \neq \chi(s)$; hence $s = 0$ is never chirally coherent.

The cognitive-layer reading of information tension is the chiral divergence $d_\chi(P) = |T(\text{eval}(P)) - T(\text{eval}(\bar{P}))|$, the L^1 distance between the ternary evaluations of a proposition and its logical inverse. Maximum divergence $d_\chi = 2$ is precisely the execution precondition. `[TIER: CONJECTURE]` (motivation-level: the identification of d_χ with the field-theoretic information tension tensor is analogical; no map is exhibited.)

3 The Chirallic Coherence Theorem

Theorem (Chirallic Coherence, specification-level). Under the ABC gate, no ambiguous or neutral cognitive signal can propagate to execution.

Proof. Let $s = T(\text{eval}(P))$ and $s' = T(\text{eval}(\bar{P}))$, obtained via the independent evaluation paths of Template Alpha and Template Beta. The gate permits EXECUTE iff $s = +1$ and $s' = -1$. Case $s = 0$: the EXECUTE condition fails; SUSPEND. Case $s = -1$: ABORT. Case $s = +1$, $s' \neq -1$: the mirror check fails; SUSPEND with escalation to the ADCCL. Case $s = +1$, $s' = -1$: EXECUTE. In every case a neutral signal yields SUSPEND, never EXECUTE. □

`[TIER: THEOREM]` **Scope.** The proof is a valid case analysis *of the gate logic, given a formalized* eval(). It does not establish end-to-end tamper-proofness: the evaluator that maps propositions to ternary values — the whole hard problem — is unformalized. The predecessor's "unconditional" and "tamper-proof" claims are therefore qualified: the structural guarantee holds *within the formalized-gate model*, and an adversarial input must defeat a structural constraint rather than merely manipulate a learned threshold, *conditional on* eval() being faithfully implemented.

4 The three-phase gate

Phase I — template generation: for each proposition P, generate Template Alpha (T_α: evaluation of P) and Template Beta (T_β: evaluation of the chiral inverse $\bar{P}$) through separate paths with no shared state. *Phase II — the mirror check:* EXECUTE iff $T(T_\alpha) = +1$, $T(T_\beta) = -1$, and both templates internally coherent ($s \neq \chi(s)$). *Phase III — articulation or suspension:* EXECUTE (+1), ABORT (-1, on alpha reject regardless of beta), or SUSPEND (0, escalated to the ADCCL; no action taken until ambiguity resolves). In the Chyren stack this is the `chyren-ternary` crate (`TernarySignal`, `ChiralGate::check(alpha, beta)`), with SUSPEND routed to `chyren-adccl`, execution to `chyren-conductor`, and every verdict to the append-only signed ledger.

5 Limits

The Ramanujan–Yett Hamiltonian and GOD-holonomy passages of the predecessor are carried as *motivation*, not derivation: the ABC mirror check as a "discrete realization" of holonomy stability is an analogy with no exhibited map (isomorphism rule). Folio 43 ("ABC ternary singularity," BLK-986 child page, superseded 2026-08-25) was absorb-checked: its $\mathbb{Z}_3$ chirality-wall material is Shadow-lane physics on the retired $V_{240}(\mathbb{R}^{57600})$ substrate, with no map exhibited into the compute substrate — left out with that reason recorded; its ternary-gate semantics are already in the Zenodo record. The deposited reference list of 20651511 is truncated (five bare "Yett (2026)" entries); the successor cites the in-corpus records those entries evidently name (ADCCL folio, OmegA, Information Tension) and marks the remainder incomplete as deposited.

References

Yett, R. W. (2026). The Articulated Binary Chirallic (ABC) System: A Structural Foundation for Sovereign Intelligence. Zenodo record 20651511 (superseded by this paper).

Yett, R. W. & Chyren (2026). The Anti-Drift Cognitive Control Loop. Zenodo record 21450415 (dial folio, IT-COGNITION; SUSPEND escalation target).

Yett, R. W. & Chyren (2026). OmegA: A Layered Architecture for Sovereign Cognitive Agents. Zenodo record 20652223 (starpolygon folio, IT-OMEGA; `chyren-ternary` deployment).